

Eco 202 - Fall 2026

Class 07

$$n=6 \Rightarrow N_0 = \# \text{ of control} = 3$$

$$N_1 = \# \text{ of treat} = 3$$

control treat

$$(\underbrace{0, 0, 0}_{\text{control}}, \underbrace{1, 1, 1}_{\text{treat}}) = (D_1, D_2, D_3, D_4, D_5, D_6)$$

Assign 1 :

Assg 2 : $(\underline{0}, \underline{0}, \underline{1}, \underline{0}, \underline{1}, \underline{1})$

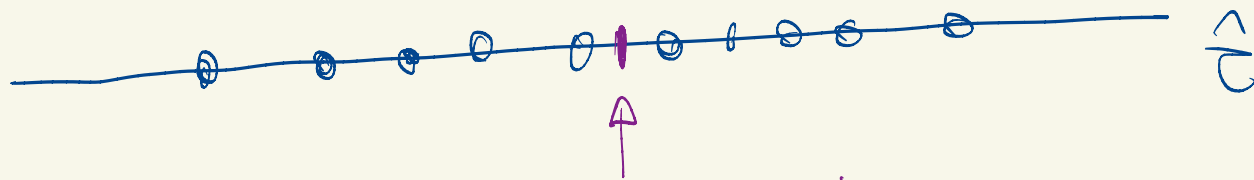
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Assig 20 : $(1, 1, 1, 0, 0, 0)$

• Complete Randomization $\Rightarrow$ all 20 assignments are equally likely -

• Each of them has chance : $\frac{1}{20}$

- $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ is random. We have no possible $\hat{\tau}$'s because we have no possible treatment assignments.



$$\underline{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))$$

Population
 $N = 1,000$

Sample
 $n = 200$

Control
 $N_0 = 100$

Treatment
 $N_1 = 100$

Random Sampling

$Y_i = \text{outcomes}$

$D_i = \text{treatment assignment}$

$i = 1, 2, \dots, n \geq 200$
(e.g. SRS)

Random assignment

treatment
assignment
mechanism

(e.g. complete
randomization)